

Session:17

1. math module: sqrt, factorial, pi

```
import math

# Example numbers
num_sqrt = 25
num_fact = 5

print("Square root:", math.sqrt(num_sqrt))
print("Factorial:", math.factorial(num_fact))
print("Value of pi:", math.pi)
```

2. List .jpg and .png files using os module

```
import os

files = os.listdir()

image_files = [f for f in files if f.lower().endswith(('.jpg', '.png'))]

print("Image files:")
for file in image_files:
    print(file)
```

3. Day of the week from date input

```
from datetime import datetime

date_str = input("Enter date (YYYY-MM-DD): ")

date_obj = datetime.strptime(date_str, "%Y-%m-%d")

print("Day of the week:", date_obj.strftime("%A"))
```

4. Custom module: insta_utils.py

Step 1: Create `insta_utils.py`

```
def format_follower_count(n):
    if n >= 1_000_000:
        return f"{n/1_000_000:.1f}M"
    elif n >= 1_000:
        return f"{n/1_000:.1f}K"
```

```
else:
    return str(n)
```

Step 2: Use it in another script (e.g., main.py)

```
import insta_utils

counts = [1500, 2300000, 987]

for c in counts:
    print(insta_utils.format_follower_count(c))
```

5. Virtual environment + statistics.mean()

Step 1: Create and activate virtual environment

Windows:

```
python -m venv venv
venv\Scripts\activate
```

Mac/Linux:

```
python3 -m venv venv
source venv/bin/activate
```

Step 2: Install packages

```
pip install statistics requests
```

Note: statistics is actually built-in in Python, so it usually does **not need installation**.

Step 3: Python script using statistics.mean()

```
import statistics

numbers = [10, 20, 30, 40, 50]

avg = statistics.mean(numbers)

print("Average:", avg)
```